14 Normal Modes pt 2

We have a matrix representing the normal modes of our system

$$\mathbb{N}(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} = \begin{pmatrix} A_+ \cos(\omega_+ t) \\ A_- \cos(\omega_- t) \end{pmatrix}$$

It is nice to express this as a complex function, but where we could still recover this. Cos is the real component of $e^{i\omega t}$, so we write the complex mode matrix

$$\mathbb{N}' = \begin{pmatrix} C_+ e^{i\omega_+ t} \\ C_- e^{i\omega_- t} \end{pmatrix}$$

Where $C_+ = A_+ e^{i\delta t}, \text{ and } C_- = A_+ e^{i\delta t}$.

We rewrite $\mathbb{N}'(t)$ as

$$\mathbb{N}'(t) = C_+ \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{i\omega_+ t} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{i\omega_- t}$$

where these functions are clearly just the basis functions.
For our original variables, we can write out $\mathbb{Z}(t)$.

$$\mathbb{Z}(t) = \begin{pmatrix} z_1(t) \\ z_2(t) \end{pmatrix} = \frac{C_+}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{i\omega_+ t} + \frac{C_-}{2} \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{i\omega_- t}$$

To go back to our actual values, we just have

$$\mathbb{X}(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} = \mathrm{Re}(\mathbb{Z}(t))$$

What do we do when we can't guess the normal modes?

Matrix approach

Our goal is to write the E-L equations as a vector equation, and solve for the eigenvectors.

Lets take our E-L equations from before

$$\begin{gathered} \ddot{x}_1 = -kx_1 - k'x_1 + k'x_2 \\ \ddot{x}_2 = -kx_2 - k'x_2 + k'x_1 \\ m \begin{pmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{pmatrix} = -\begin{pmatrix} k + k' - k' \\ -k' + k + k' \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \end{gathered}$$

We can rewrite this in a nicer matrix way

$$\ddot{\mathbb{X}} - \frac{1}{m} \mathbb{K} \mathbb{X}$$

We can solve this by guessing through an Ansatz.
Lets look at the slightly more complicated but nicer matrix $\mathbb{Z}$.

$$\ddot{\mathbb{Z}} = -\frac{1}{m} \mathbb{K} \mathbb{Z}$$

With the Ansatz $\mathbb{Z}(t) = \underbrace{\mathbb{Z}_0}_{\text{constant from init conds}} e^{i\omega t}$

We have

$$
\begin{aligned}
\frac{d^2}{dt^2}(\mathbb{Z}_0 e^{i\omega t}) &= -\frac{1}{m}\mathbb{K}\mathbb{Z}_0 e^{i\omega t} \\
-\omega^2 \mathbb{Z}_0 e^{i\omega t} &= -\frac{1}{m}\mathbb{K}\mathbb{Z}_0 e^{i\omega t} \\
m\omega^2 \mathbb{Z}_0 &= \mathbb{K}\mathbb{Z}_0
\end{aligned}
$$

This is an eigenvalue (or eigenmode) equation.
Note that we have different ω for each normal mode.

$$
\begin{aligned}
x_+ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \omega_+ &= \sqrt{\frac{k}{m}} \\
x_- \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \omega_- &= \sqrt{\frac{k+2k'}{m}}
\end{aligned}
$$

We can solve the eigenvalue problem

$$
\begin{aligned}
\mathbb{K}\mathbb{Z}_0 - m\omega^2\mathbb{Z}_0 &= 0 \\
(\underbrace{\mathbb{K} - m\omega^2}\ )\mathbb{Z}_0 &= 0
\end{aligned}
$$

This is of the form $\mathbb{Z}_0 = 0$
We can solve this with

$$
\begin{aligned}
{}^{-1}\mathbb{Z}_0 &= {}^{-1}0 \\
\mathbb{Z}_0 &= \begin{pmatrix} 0 \\ 0 \end{pmatrix}
\end{aligned}
$$

We need a non invertible matrix

$$
\det(\mathbb{K} - m\omega^2) = 0
$$

Lets find the eigenvalues

$$
\begin{pmatrix} k + k' - m\omega^2 & -k' \\ -k' & k + k' - m\omega^2 \end{pmatrix} = 0
$$

Taking the determinant yields

$$
(k + k' - m\omega^2)^2 - k'^2 = 0
$$

This is a difference of two squares, in the form $(a+b)(a-b)$

$$
(k + k' - m\omega^2 + k')(k + k' - m\omega^2 - k') = 0
$$

This yields two different solutions for what ω can be (which should agree with what we found before)!

$$
\begin{aligned}
m\omega^2 &= k + 2k' \\
\omega_1 = \sqrt{\frac{k+2k'}{m}} &= \omega_-
\end{aligned}
$$

We can plug in the values of ω to find for our eigenvectors.

For example

$$
\begin{aligned}
(\mathbb{K} - m\omega^2)\begin{pmatrix} a \\ b \end{pmatrix} &= 0 \\
\begin{pmatrix} k\ + k' - \frac{m\ (\ k\ + 2k')}{m} & -k \\ -k' & -k' \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} &= \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\
k' \begin{pmatrix} -1 & -1 \\ -1 & -1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} &= \begin{pmatrix} 0 \\ 0 \end{pmatrix}
\end{aligned}
$$

This gives us the equations
$-a-b=0$, so $a=-b$.

This is our eigenvector, $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$

You can do the same thing with the other normal mode for $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

We now have the complex functions of time with any complex coefficients A_1, A_2 as

$$\mathbb{Z}(t) = A_1 \mathbb{Z}_0^1 e^{i\omega_1 t} + A_2 \mathbb{Z}_0^2 e^{i\omega_2 t}$$
$$\text{here here,}$$
$$\mathbb{Z}_0^1 = \frac{1}{2}\begin{pmatrix} 1 \\ -1 \end{pmatrix}, \omega_1 = \sqrt{\frac{k+2k'}{m}}$$
$$\mathbb{Z}_0^2 = \frac{1}{2}\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \omega_2 = \sqrt{\frac{k}{m}}$$

(note that the $\frac{1}{2}$ here is arbitrary, but we write it to match)

Practice problem

Lets take three masses with 4 springs between them, all of mass m and spring constant k.

The legrangian for this is

$$U = \frac{1}{2}k(x_3^2 + (x_3 - x_2)^2 + (x_2 - x_1)^2 + x_1^2)$$
$$T = \frac{1}{2}m(\dot{x_1}^2 + \dot{x_2}^2 + \dot{x_3}^2)$$
$$\mathscr{L} = \frac{1}{2}m(\dot{x_1}^2 + \dot{x_2}^2 + \dot{x_3}^2) - \frac{1}{2}k(x_3^2 + (x_3 - x_2)^2 + (x_2 - x_1)^2 + x_1^2)$$

We can find the EL equations for these

$$m\ddot{x}_1 = \frac{1}{2}k(2x_1 - 2x_2 + 2x_1)$$
$$m\ddot{x}_2 = \frac{1}{2}k(2x_2 - 2x_3)$$
$$m\ddot{x}_3 = \frac{1}{2}k(2x_3 + 2x_3 - 2x_2)$$

We can write these as a matrix form

$$\begin{pmatrix} \ddot{x_1} \\ \ddot{x_2} \\ \ddot{x_3} \end{pmatrix} = \frac{k}{2m}\begin{pmatrix} 4 & -2 & 0 \\ 0 & 2 & -2 \\ 0 & -2 & 4 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

We can take this as the matrix equation

$$\ddot{\mathbb{X}} = \frac{k}{2m}\mathbb{K}$$